Orbit intersections and regular semidirect geometries

Human Compression Project

Abstract

An action on the types of a regular geometry can merge several types into one. The relevant condition is an intersection identity for the orbits of its parabolic subgroups. We prove the finite-rank twisting theorem directly from this identity, giving the stabilizer of every flag and the full type-preserving automorphism group. A second theorem obtains the identity from an action on the elements of another flag-transitive geometry. Together with an orbital Coxeter presentation, these statements supply both constructions in Piedade–Tranchida's paper on j-diagonals.

1 Conventions and the two intersection properties

All type sets in this paper are finite. Cosets are left cosets. In a coset geometry $(B, (B_i)_{i\in I})$ write $B_J = \bigcap_{i\in J} B_i$, with $B_\varnothing = B$. A regular hypertope is thin, residually connected and flag-transitive. Automorphisms preserve types. In its faithful standard model, the group acts freely and transitively on chambers. Its distinguished involutions b_i interchange a fixed chamber and its i-adjacent chamber; then

$$B_J = \langle b_i : i \in I \setminus J\rangle, \qquad B_I = 1. \tag{1}$$

We recall why the geometric hypotheses give these assertions. In a thin geometry a type-preserving automorphism fixing a chamber fixes all its adjacent chambers. Residual connectivity implies that the chamber graph, and the chamber graph of every residue, is connected: by induction on rank, chambers containing a given element are connected; a path in the incidence graph connects these families, using chambers containing each consecutive incident pair. The rank-one and rank-two cases start the induction. Thus a chamber stabilizer is trivial. The same argument inside the residue of the base flag of types J proves (1). It also proves that the full type-preserving automorphism group is the displayed group whenever a faithful group already acts transitively on chambers.

For a second regular hypertope $(A, (A_x)_{x\in S})$, put $A(T) = \langle a_x : x \in T\rangle$ for $T \subseteq S$. Formula (1) gives

$$A(T) \cap A(T') = A(T \cap T'). \tag{2}$$

We will use both this intersection property and flag-transitivity. They are different requirements: the intersection property alone is not a general criterion for a nonstring coset system to be a geometry.

Lemma 1 (Coset intersection)**.** *If B is flag-transitive on its coset geometry, every finite pairwise-intersecting family of cosets of standard parabolic subgroups has nonempty total intersection. In particular, for $H, K, J \subseteq I$,*

$$B_H B_J \cap B_K B_J = B_{H\cup K} B_J.$$

Proof. Expand a coset gB_H as $\bigcap_{i\in H} gB_i$. Pairwise intersection of the original cosets implies pairwise intersection of all the resulting maximal-parabolic cosets. Intersecting cosets of the same type are equal. They therefore form a flag, which flag-transitivity takes to a standard flag; its cosets have a common group element. For the last assertion, if $g \in B_H B_J \cap B_K B_J$, the three cosets B_H, B_K, gB_J intersect pairwise. A point of their common intersection belongs to $B_{H\cup K}$ and to gB_J. The reverse inclusion follows from subgroup containment. □

Theorem 2 (Element orbits)**.** *Fix $j \in I$ and $x_0 = B_j \in X = B/B_j$. If B is flag-transitive, then*

$$(B_H x_0) \cap (B_K x_0) = B_{H \cup K} x_0 \qquad (H, K \subseteq I). \tag{3}$$

Thus the complementary notation $O(M) = B_{I \setminus M} x_0$ satisfies $O(M) \cap O(N) = O(M \cap N)$.

Proof. The inverse image of $B_H x_0$ under $B \to B/B_j$ is $B_H B_j$. Apply Lemma 1 and take images. Finally $(I \setminus M) \cup (I \setminus N) = I \setminus (M \cap N)$. □

2 The common twisting theorem

Let $\alpha = (A, (A_x)_{x \in S})$ and $\beta = (B, (B_i)_{i \in I})$ be regular hypertopes in their faithful standard models. Suppose B acts on A by automorphisms permuting the distinguished generators a_x and their types. Let $\mathcal{K}$ be the set of orbits on S, and choose $x_L \in L$ for each $L \in \mathcal{K}$. Assume

$$B_H x_L \cap B_K x_L = B_{H \cup K} x_L \quad (H, K \subseteq I,\ L \in \mathcal{K}). \tag{4}$$

The notation $A(T)$ always denotes the subgroup generated by T; the source notation A_L for an orbit L instead denotes the intersection of the maximal parabolics indexed by all elements of L, namely $A(S \setminus L)$.

Set $G = A \rtimes B$ and $T = I \sqcup \mathcal{K}$. Define

$$G_i = A\left(\bigcup_{L \in \mathcal{K}} B_i x_L\right) \rtimes B_i \ (i \in I), \qquad G_L = A(S \setminus L) \rtimes B \ (L \in \mathcal{K}). \tag{5}$$

The supports displayed here are invariant under the respective second factors, so these are subgroups.

Theorem 3 (Regular semidirect geometry)**.** *The coset system $(G, (G_t)_{t \in T})$ is a regular hypertope, with full type-preserving automorphism group G and distinguished involutions*

$$\{b_i : i \in I\} \cup \{a_{x_L} : L \in \mathcal{K}\}.$$

For $J \subseteq T$, write $J_B = J \cap I$, $J_A = J \cap \mathcal{K}$ and

$$S(J) = \bigcup_{L \in \mathcal{K} \setminus J_A} B_{J_B} x_L.$$

Its standard flag stabilizer is exactly

$$G_J = A(S(J)) \rtimes B_{J_B}. \tag{6}$$

The geometry is finite if and only if both input geometries are finite.

Proof. Intersect the subgroups in (5) inside the Cartesian set $A \times B$. Different B-orbits in S are disjoint. Equation (4), iterated over the finite set J_B, shows that the intersection of the supports is precisely $S(J)$. Equation (2) now proves (6).

We next prove flag-transitivity, rather than infer it from the subgroup intersection formula. Project any flag of G-cosets to B. The cosets of old types project to a pairwise-intersecting family of B_i-cosets, while new types project to B. By Lemma 1 these have a common element b. Left translation by $(1, b^{-1})$ makes every projected coset contain 1. Each original coset then has a representative $(w, 1)$. As a subset it is the Cartesian product of a coset of a standard parabolic of A and the corresponding subgroup of B. Pairwise incidence therefore implies pairwise intersection of these A-cosets. Lemma 1, now for α, gives a common A-coordinate. Thus the original flag has a common group element and is an image of the standard flag. Every flag extends to a chamber.

Equation (6) gives $G_T = 1$. If only an old type i is missing from J, it gives $G_J = \langle b_i \rangle$; if only an orbit type L is missing, it gives $G_J = \langle a_{x_L} \rangle$. Every panel consequently has two chambers, proving thinness.

We verify residual connectivity in full. For $|T \setminus J| \geq 2$, choose two distinct missing types u, v. We claim

$$G_J = \langle G_{J\cup\{u\}}, G_{J\cup\{v\}} \rangle. \tag{7}$$

If both are old types, their B-parts generate B_{J_B} by (1). For each orbit L not deleted by J_A, either smaller support contains x_L, since every subgroup orbit contains its base point. Conjugating these generators by B_{J_B} recovers every a_x with $x \in S(J)$. If one missing type is old and the other is an orbit L, the subgroup obtained by deleting L already contains B_{J_B} and the full support from every other remaining orbit; the old-type subgroup contains a_{x_L}, whose B_{J_B}-conjugates recover the remaining orbit. If both missing types are orbits, both subgroups contain B_{J_B} and the union of their supports is $S(J)$. This proves (7) in all cases.

The residue of the standard flag is the coset geometry for G_J with parabolics $G_{J\cup\{t\}}$. Indeed, flag-transitivity identifies every residue element with such a coset. A coset incidence graph is connected when its defining subgroups generate its group: multiplication by a factor from one defining subgroup joins the corresponding incident cosets, and a word in these factors gives a path. Equation (7) applies to every residue of rank at least two, so all are connected. Chamber propagation from the first section proves both faithfulness and the full automorphism assertion. Finally a finite-rank faithful regular geometry has finitely many elements exactly when its chamber group is finite. The semidirect product is finite exactly when A and B are finite. □

This is the finite-rank regular case of Piedade–Tranchida's generalized twisting theorem [2, Corollary 3.17]. The proof here includes its flag-transitivity, residue-generation and thinness arguments; that corollary is therefore an attribution, not an unproved premise. The source's more general incidence systems, infinite-rank products, wreath constructions and later examples are not recovered by this theorem.

3 Invariant quotients and orbital presentations

Lemma 4 (Quotients)**.** *If $M \leq A$ is invariant under B, its normal closure N in A is invariant. The action descends to A/N and permutes the images of the standard maximal parabolics with the same indexed type action. If their coset geometry is a regular hypertope, the corresponding twisting is again a regular hypertope.*

Proof. The normal closure is generated by ama^{-1} with $a \in A, m \in M$. An automorphism from B carries each such generator to another of the same form, and its inverse gives equality. The quotient action and parabolic permutation follow. Orbit identity (4) concerns only the indexed types and is unchanged.

For completeness, the regular-geometry assumption can be interpreted without silently assuming a faithful quotient action. Put $\bar{A} = A/N$, $C_x = A_x N/N$, and $C_E = \bigcap_{x\in E} C_x$, with $C_\varnothing = \bar{A}$. The source's quotient twisting uses intersections of these maximal parabolics:

$$\bar{G}_i = C_{S\setminus\bigcup_L B_i x_L} \rtimes B_i \quad (i \in I), \qquad \bar{G}_L = C_L \rtimes B \quad (L \in \mathcal{K}). \tag{8}$$

These are not defined by substituting the images of generated-support subgroups into (5).

Each image of a_x fixes the base flag with type x deleted. It cannot belong to C_x: otherwise C_x, already containing the other generators, would be all of $\bar{A}$, giving only one element of type x, contrary to thinness. It therefore performs the unique adjacency of that type. Chamber-graph connectivity makes $\bar{A}$ flag-transitive. Its chamber stabilizer $H = C_S$ fixes every chamber by thinness, so is its normal action kernel. It is B-invariant because it is the intersection of the permuted C_x.

Every C_E contains H, and intersections commute with passage to $\bar{A}/H$. In that faithful regular model formula (1) therefore gives

$$C_E/H = \bigcap_{x\in E}(C_x/H) = \langle a_x H : x \in S \setminus E\rangle.$$

Consequently every subgroup in (8) is the full inverse image of the corresponding subgroup from Theorem 3 for $\bar{A}/H$. Apply that theorem. Passage through the normal subgroup $H \rtimes 1$ preserves all cosets and their intersections, so the original quotient twisting has the same regular geometry. For a transitive type orbit $L = S$, its new parabolic is $H \rtimes B$, not necessarily B. The quotient group $\bar{A}$ itself is not asserted to be the faithful automorphism group. □

Proposition 5 (Orbital presentation)**.** *Let B act transitively on a finite set X with base point x_0 and stabilizer P. Give each orbit of unordered distinct pairs a Coxeter label. Let Y be a finite set fixed by B, with fixed labels within Y and one label m_y between y and every element of X. Put $A = W(Y \sqcup X)$. Then $A \rtimes B$ has the presentation consisting of a presentation for B, one involution $a = a_{x_0}$, the involutions c_y for $y \in Y$, and:*

1. *$[a, p] = 1$ for generators p of P, and $[c_y, b] = 1$ for generators b of B;*
2. *the prescribed relations among the c_y, and $(c_y a)^{m_y} = 1$ when $m_y < \infty$;*
3. *$(a\tau a\tau^{-1})^m = 1$ for one representative $\{x_0, \tau x_0\}$ of each unordered pair orbit of finite label m.*

The unordered orbits are indexed by the inverse-paired double cosets $P\tau P \cup P\tau^{-1}P$ with $\tau \notin P$.

Proof. In the proposed group define $a_{bx_0} = bab^{-1}$. The stabilizer relations make this independent of b. Double-coset multiplication changes the representative without changing its pair orbit, and inversion interchanges its endpoints. Conversely, a transporter of two unordered pairs either fixes the selected first endpoint or interchanges the endpoints, giving exactly the two double cosets displayed. Hence conjugates of the chosen relations give every Coxeter relation on X. Conjugating the relations with c_y, which is fixed by B, gives all the mixed Coxeter relations. These elements therefore define a homomorphism from $W(Y \sqcup X) \rtimes B$ onto the proposed group. The reverse map takes the proposed generators to their named elements in the semidirect product; all the relators hold there. Both composites fix the generating set, proving the isomorphism. This is a presentation of the full group, not merely a finite image. □

Lemma 6 (Exact doubled label)**.** *In $A \rtimes B$ let $a = a_x$ and let $b \in B$ be an involution with $bx \neq x$. If $a_x a_{bx}$ has order m (possibly infinite), then ab has order $2m$, with $2\infty = \infty$.*

Proof. We have $(ab)^2 = a_x a_{bx}$. Projection to B sends ab to the nontrivial involution b, so any power of ab equal to 1 has even exponent. For an even exponent $2k$ it is 1 precisely when $(a_x a_{bx})^k = 1$. This proves exactness in both the finite and infinite cases. □

4 Attribution and retained facts

The constructions and target results are due to Piedade and Tranchida [1,2]. The orbital presentation is the same argument used for the matching extensions in our earlier common paper, with the fixed vertices included explicitly. The contribution of this proof edition is a single complete regular semidirect theorem and explicit adapters, not a priority claim for generalized twisting. The classical Coxeter facts used by the application are that standard parabolic subgroups embed with their indicated Coxeter presentations and that the standard coset geometry is a regular hypertope. The usual string C-group–regular-polytope correspondence and the named classical twisting constructions are retained as stated in the application. The companion Lean file verifies only the declared coset, support, and order identities; it is not an end-to-end geometry formalization.

References

[1] C. A. Piedade and P. Tranchida, *Coset geometries acting on their elements: The j-diagonals twisting*, arXiv:2608.23223v1, 2026.

[2] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 2026, Definition 3.10 and Corollary 3.17.